

Classical Chinese OCR and Sentence Segmentation for Vietnamese Historical Documents

Nguyen Phan Khanh Linh — 25C01010

Lam Minh Quang — 25C01018

Nguyen Vo Hoang Phuc — 25C01042

Faculty of Mathematics and Computer Science

University of Science, Vietnam National University Ho Chi Minh City

Natural Language Processing

Lecturer: Assoc. Prof. Dr. Dinh Dien

1 Introduction

1.1 Project Objective

Vietnamese historical documents written in Classical Chinese constitute an important source for research on the history, administration, diplomacy, and intellectual traditions of Vietnam. However, many of these materials remain available only as scanned or photographed page images. Although image-based archives preserve the visual form of the documents, they provide limited support for full-text search, corpus analysis, and natural language processing.

The objective of Topic 9 is to construct a monolingual Classical Chinese corpus for Vietnamese historical studies. The group develops an automated pipeline that converts historical document images into structured textual data through two main stages: optical character recognition (OCR) and sentence segmentation.

The official pipeline uses PaddleOCR 3.7.0 to recognize Classical Chinese characters from page images. The resulting OCR text is then processed by a Classical Chinese punctuation model hosted on Hugging Face to restore punctuation and identify sentence boundaries. The final corpus is produced exclusively by these two components.

1.2 Assigned Task

Because the assigned input consists of document images, the project requires the following tasks:

1. **Optical character recognition:** detect and recognize the Classical Chinese text contained in each page image;
2. **Sentence segmentation:** restore punctuation and divide the OCR text into individual sentence units.

For each historical work, the pipeline produces two required files:

- `[document_id]_raw.txt`, containing the reconstructed raw OCR text;
- `[document_id]_seg.tsv`, containing one segmented sentence per line in the format `sentence_id<TAB>sentence`.

The sentence-segmentation stage operates directly on the corresponding OCR output. It restores punctuation and determines sentence boundaries without translating, paraphrasing, or intentionally correcting the recognized Classical Chinese characters.

1.3 Data Scope and Expected Outcomes

The working dataset contains 33 historical works and 2,180 page images. All works are processed using the same pipeline to maintain consistency in page order, OCR generation, sentence segmentation, file naming, and output organization.

The expected outcomes are:

- 33 raw OCR text files, one for each historical work;
- 33 sentence-segmented TSV files following the required format;
- consistent mapping between source identifiers and the required HVH corpus identifiers;
- structured data suitable for storage, retrieval, and later linguistic or computational analysis;
- document-level similarity results used to assess the consistency of the official pipeline.

Because no complete expert-verified transcription or sentence annotation is available, the project does not claim absolute OCR or segmentation accuracy. Instead, independent reference outputs from Gemini and Gemma are used to provide additional evidence about the consistency of the official results.

2 Data and Tools Used

2.1 Input Data Description

The input data consist of scanned or photographed pages from Vietnamese historical works written in Classical Chinese. The collection primarily contains traditional Chinese characters and historical glyph variants arranged in vertical writing layouts. Many source pages contain dense text and limited or inconsistent punctuation.

Image quality varies across the dataset. Common difficulties include low contrast, faded ink, page skew, shadows, border noise, uneven illumination, damaged characters, and differences in image resolution. These conditions may affect text-region detection, character recognition, and reading-order reconstruction.

Sentence segmentation is also challenging because Classical Chinese texts often contain little or no explicit punctuation. Sentence boundaries must therefore be inferred from contextual and syntactic information rather than detected only from visible punctuation marks.

The images are grouped by historical work and arranged in their original page order. Preserving this order is essential because missing, duplicated, or incorrectly ordered pages would alter the reconstructed OCR sequence and affect the segmentation results.

2.2 Data Source

The dataset was provided for Topic 9 of the Natural Language Processing course. It contains 33 Vietnamese historical works written in Classical Chinese, comprising a total of 2,180 page images.

The document images were collected from the Nôm Foundation digital library using the links and metadata supplied with the project assignment. The image collection is stored separately from the source-code repository because of its size and data-distribution constraints.

The project metadata are used to associate images with their corresponding historical works, verify image counts, preserve page order, and map source identifiers to the required HVH corpus identifiers.

2.3 Tools, Libraries, and Models Used

The project distinguishes between components used to produce the official corpus and components used only for evaluation. The main components are summarized in Table 1.

Table 1: Main components used in the project.

Component	Function	Role in the project
PaddleOCR 3.7.0	OCR	Recognizes Classical Chinese characters from document images and produces the official raw text.
Hugging Face punctuation model	Segmentation	Restores punctuation and determines sentence boundaries in the OCR output.
Gemini and Gemma APIs	Reference	Generate independent outputs used only for pairwise comparison with the official pipeline.
RapidFuzz, Pandas, and Matplotlib	Evaluation	Compute similarity metrics, aggregate document-level results, and produce summary visualizations.

2.3.1 PaddleOCR 3.7.0

PaddleOCR 3.7.0 is the primary OCR framework used to generate the official raw corpus. For each page image, it detects text regions and recognizes the corresponding Classical Chinese characters. The page-level outputs are then combined according to the original document order and stored in the required `raw.txt` file.

Running PaddleOCR locally allows the same configuration to be applied consistently to all 2,180 images. It also preserves the original recognition output without hidden corrections from an external commercial service.

The OCR stage is responsible only for text extraction. It does not perform translation, semantic interpretation, or manual correction of the recognized characters.

2.3.2 Classical Chinese Punctuation Model

Sentence segmentation is performed using the Hugging Face model:

`raynardj/classical-chinese-punctuation-guwen-biaodian`

The model is based on Chinese BERT and formulated as a token-classification task. It receives an unpunctuated or sparsely punctuated OCR sequence and predicts punctuation labels for individual character positions.

The pipeline inserts the predicted punctuation into the OCR text. Sentence-ending marks are then used to divide the continuous text into sentence-level records, which are written to the required `seg.tsv` file.

This contextual model is suitable for Classical Chinese documents because sentence boundaries are frequently not marked explicitly in the original sources.

2.3.3 Reference Comparison

Besides the official PaddleOCR pipeline, we also investigated an alternative OCR approach based on Large Language Models (LLMs), specifically Google’s Gemini and Gemma models. Instead of using a conventional OCR engine, these models receive document images together with a carefully designed prompt that instructs them to transcribe only the visible Classical Chinese characters while preserving the original reading order and avoiding translation or interpretation.

The prompt explicitly requires the models to:

- transcribe only characters that are clearly visible in the image;
- preserve the original vertical reading order (top-to-bottom, right-to-left);
- avoid translation, explanation, or character normalization;
- stop rather than guess when the text cannot be recognized confidently;
- return only the extracted text without any additional formatting.

This approach demonstrates several advantages. For document pages with good image quality, clear printing, and relatively simple character structures, Gemini and Gemma often produce highly accurate transcriptions. Their strong visual-language reasoning capability also enables them to handle certain layout variations that may challenge traditional OCR systems.

However, the limitations become apparent when processing degraded historical documents. Images containing faded ink, blurred strokes, damaged characters, or densely written complex Chinese characters frequently cause the language models to generate characters that are not actually present in the source image. In other words, the models may hallucinate plausible-looking Classical Chinese characters instead of faithfully reproducing the original text. Such behavior reduces transcription reliability and makes the outputs unsuitable for constructing an authoritative historical corpus without further verification.

For this reason, Gemini and Gemma are not used to generate the official corpus. Instead, their outputs serve only as independent reference transcriptions for comparison with the PaddleOCR–Hugging Face pipeline. Pairwise similarity between the different OCR outputs is evaluated using normalized Levenshtein similarity after basic text normalization.

Gemini and Gemma APIs are used only to generate two independent reference outputs. They do not contribute text to the official submitted corpus and are not treated as ground truth.

The three outputs form the following comparison pairs:

- PaddleOCR–Hugging Face and Gemini;
- PaddleOCR–Hugging Face and Gemma;
- Gemini and Gemma.

For OCR, pairwise agreement is measured using normalized Levenshtein similarity after basic text normalization. This metric reflects differences caused by character insertion, deletion, or substitution.

2.3.4 Supporting Libraries

RapidFuzz is used for normalized Levenshtein similarity and character-sequence alignment. Pandas organizes document-level measurements and exports CSV results, while Matplotlib produces summary visualizations.

Pillow and OpenCV support image loading and optional preprocessing. Transformers, SentencePiece, and Safetensors support model loading and inference, while NumPy and tqdm provide numerical operations and progress monitoring. These packages support implementation, but the two central processing components remain PaddleOCR 3.7.0 and the Hugging Face punctuation model.

3 Processing Workflow

3.1 Overall Workflow

The project implements an end-to-end pipeline for constructing a monolingual Classical Chinese corpus from historical document images. The official processing path consists of image organization, optional preprocessing, OCR, reading-order reconstruction, punctuation restoration, sentence segmentation, and output validation.

The main workflow is:

Document images

- > optional preprocessing
- > PaddleOCR 3.7.0
- > document-level raw.txt
- > Classical Chinese punctuation model
- > document-level seg.tsv
- > format and content validation

PaddleOCR 3.7.0 and the Hugging Face punctuation model form the official corpus-generation pipeline. Gemini and Gemma are processed separately and used only as supplementary reference systems. Their outputs do not modify or replace the official corpus.

3.2 Main Processing Steps

3.2.1 Image Organization and Preprocessing

The 2,180 page images are grouped into 33 document directories. Images within each directory are sorted by filename to preserve their original page order. This step is necessary because incorrect ordering would alter the reconstructed document even when the page-level OCR results were individually correct.

Optional preprocessing is implemented with OpenCV. Images may be converted to grayscale, enhanced using Contrast Limited Adaptive Histogram Equalization, and deskewed when a noticeable rotation is detected. Stronger denoising and adaptive thresholding are reserved for severely degraded pages because excessive preprocessing may remove thin character strokes.

The original image is used whenever its quality is sufficient. This conservative approach avoids unnecessary transformations and preserves the visual details required for recognizing complex historical characters.

3.2.2 Optical Character Recognition

OCR is performed using PaddleOCR 3.7.0 with the traditional Chinese recognition configuration:

```
lang = chinese_cht
```

Text-line orientation support is enabled to handle rotated and vertical regions. Document orientation classification and document unwarping are disabled because the input pages are already scanned or photographed as complete documents.

For each page, PaddleOCR returns recognized text, confidence values, and bounding boxes. The center coordinates of the bounding boxes are used to reconstruct the reading order.

Classical Chinese historical pages are commonly written vertically and read from right to left. Therefore, detected regions are sorted by decreasing horizontal position and increasing vertical

position. The rightmost column is processed first, while the contents of each column are ordered from top to bottom.

This reading-order adjustment is an important domain-specific component of the pipeline. Without it, correctly recognized regions could still be merged into an incorrect sequence.

Each page is first saved as an individual OCR text file. After all pages of a document have been processed, the page-level results are merged in order and stored as:

```
[document_id]_raw.txt
```

Page markers are retained in the raw file to support traceability and page-level inspection. Existing page results are skipped when the pipeline is restarted, allowing interrupted processing to resume without repeating completed work.

3.2.3 Punctuation Restoration and Sentence Segmentation

Sentence segmentation is performed using:

```
raynardj/classical-chinese-punctuation-guwen-biaodian
```

The model is loaded through the Hugging Face Transformers library using `AutoTokenizer` and `AutoModelForTokenClassification`. It predicts punctuation labels at character positions in an unpunctuated or sparsely punctuated Classical Chinese sequence.

Before inference, page-marker lines and whitespace are removed from the OCR output. Because transformer models have a limited input length, each document is divided into chunks of at most 480 characters. This value leaves space for the model’s special tokens within the 512-token limit.

Predicted punctuation marks are inserted after their corresponding characters. The punctuated text is then split using the principal sentence-ending marks:

Each sentence receives a deterministic identifier and is written in the required format:

```
sentence_id<TAB>sentence
```

The final file is saved as:

```
[document_id]_seg.tsv
```

The segmentation stage is intended only to insert punctuation and sentence boundaries. It does not deliberately translate, paraphrase, or correct the OCR characters.

3.2.4 Output Validation

Before submission, the pipeline checks that every document contains both required output files. Each non-empty TSV line must contain a sentence identifier, a tab character, and a sentence.

The character sequence reconstructed from `seg.tsv` is also compared with the corresponding `raw.txt` after page markers, identifiers, whitespace, and punctuation are removed. This check verifies that sentence segmentation has not silently rewritten the OCR content.

Source document identifiers are subsequently mapped to the required HVH corpus identifiers using the project metadata.

3.2.5 Supplementary API References

Gemini 2.5 Flash and Gemma 4 31B were explored as independent references. EasyOCR was used to identify candidate text regions, after which the prepared images were submitted to the APIs using restrictive prompts.

The prompts required the APIs to preserve the original character forms, follow the vertical right-to-left reading order, avoid translation or explanation, and insert punctuation without adding new content.

The returned outputs were converted to the same `raw.txt` and `seg.tsv` structures as the official pipeline. They were used only for supplementary comparison because generative systems may normalize characters, omit uncertain regions, or infer characters that are not clearly visible.

4 Results

4.1 Processed Data Volume

The official pipeline processed 33 document units containing 2,180 page images. Every document produced one raw OCR file and one sentence-segmented TSV file.

The official result set therefore contains:

- 33 `raw.txt` files;
- 33 `seg.tsv` files;
- page-level OCR files used for inspection and resumption;
- metadata connecting source identifiers with HVH identifiers.

The Gemini and Gemma outputs and the evaluation reports are stored separately and are not counted as official corpus deliverables.

4.2 Generated Outputs

The output structure for each document is:

```
outputs/<document_id>/
|-- <document_id>_raw.txt
'-- <document_id>_seg.tsv
```

The raw file preserves the OCR content in page order, while the TSV file provides sentence-level records suitable for corpus search and later linguistic analysis.

The project also generated supporting artifacts, including page-level OCR files, optional bounding-box visualizations, validation warnings, similarity summaries, CSV reports, and an HTML dashboard. These artifacts support inspection and evaluation but do not replace the required corpus files.

4.3 Illustrative Results

A raw OCR file retains page boundaries in the following form:

```
### nlvnpf-0250-001 ###
<recognized text of page 001>
```

```
### nlvnpf-0250-002 ###  
<recognized text of page 002>
```

The corresponding segmented output follows the required TSV structure:

```
HVH_001_000001<TAB><first sentence>  
HVH_001_000002<TAB><second sentence>
```

The two files serve complementary purposes. The raw output supports page-level traceability, while the segmented output provides structured sentence units.

5 Evaluation and Discussion

5.1 Evaluation Methodology

No complete expert-verified transcription was available for the dataset. Therefore, the project could not calculate absolute character accuracy or expert sentence-boundary accuracy for all 2,180 images.

The official pipeline was evaluated using two main forms of evidence:

1. structural and content-preservation checks within the official outputs;
2. supplementary comparison with independent Gemini and Gemma outputs.

The internal checks directly evaluate whether the pipeline produces complete, correctly formatted, and traceable outputs. The API comparison is only an exploratory consistency signal and is not treated as a benchmark or ground truth.

For OCR comparison, normalized Levenshtein similarity is computed after Unicode normalization and removal of whitespace and punctuation. The metric reflects character insertions, deletions, and substitutions between two outputs.

For sentence segmentation, Boundary F1 is calculated after the character sequences have been aligned. The metric measures whether two methods place sentence endings at comparable positions.

For summary visualization only, the two metrics are combined using equal weights:

$$S_{\text{combined}} = 0.5S_{\text{OCR}} + 0.5S_{\text{boundary}}.$$

This score is used to organize comparison results. It must not be interpreted as OCR accuracy or overall model performance.

5.2 Evaluation Results

The primary result is that the team-developed pipeline successfully processed the complete dataset and generated both required files for all 33 document units.

The OCR stage can run locally without commercial OCR quotas, preserves page-level results, supports interrupted execution, and explicitly reconstructs the right-to-left reading order of vertical Classical Chinese text. These properties make the system suitable for corpus-scale processing rather than isolated page demonstrations.

The content-preservation check provides direct evidence that the sentence-segmentation stage operates consistently with its intended function. After removing formatting elements and predicted

punctuation, the reconstructed segmented text can be compared with the raw OCR sequence to detect unintended character changes.

The supplementary comparison showed that PaddleOCR–Hugging Face appeared in the highest-ranked similarity pair for 23 of the 33 documents. This observation suggests that the official output was frequently closer to at least one independent reference than the two references were to each other.

However, the absolute similarity values were generally low and varied across documents. The values therefore cannot be used to claim that one system is correct or more accurate. They are retained only as diagnostic indicators for locating documents with unusual differences.

The principal demonstrated strengths of the official pipeline are:

- complete processing of the assigned dataset;
- local and reproducible execution;
- support for vertical right-to-left reading order;
- resumable page-level processing;
- traceable document-level outputs;
- separation of OCR from punctuation restoration;
- validation of required file structures and content preservation.

5.3 Challenges Encountered

The main difficulties were caused by faded ink, low contrast, skew, page borders, damaged strokes, and complex historical glyphs. These factors may lead to missed regions, character substitutions, or duplicated recognition.

Vertical layout also creates reading-order problems. The implemented coordinate rule works for conventional vertical columns but may be less reliable on pages containing marginal notes, seals, horizontal titles, or irregular text blocks.

Sentence segmentation remains challenging because many Classical Chinese sources contain little or no punctuation. More than one segmentation may be linguistically plausible, particularly in long historical passages.

The punctuation model also processes long documents in fixed-length chunks. Context crossing a chunk boundary is unavailable to the model and may influence punctuation decisions near that position.

Gemini and Gemma introduce additional limitations, including quotas, processing cost, output variability, and possible hallucination on unclear images. This is why they are used only as supplementary references.

5.4 Cause Analysis

OCR errors are influenced by both image quality and model coverage. PaddleOCR’s traditional Chinese model is suitable for printed traditional characters, but rare historical forms, damaged strokes, and possible Nôm characters may be underrepresented in its training data.

Image preprocessing involves a trade-off. Contrast enhancement and deskewing may improve detection, while aggressive denoising or binarization may remove fine strokes. Preprocessing is therefore applied selectively.

Reading-order errors occur when a page does not follow the assumed right-to-left column structure. More complex page-layout analysis would be required for mixed or irregular layouts.

Segmentation differences arise because punctuation restoration is a contextual linguistic task rather than a purely mechanical operation. The Hugging Face model may also differ from the project corpus in period, genre, and vocabulary.

Large language model APIs optimize plausible generation rather than strict visual transcription. Even under restrictive prompts, they may normalize characters or infer unclear content. Their disagreement with the official pipeline is therefore not direct evidence that the official result is incorrect.

6 Conclusion and Future Work

6.1 Completed Work

The project completed the two required tasks for Topic 9: OCR and sentence segmentation of Vietnamese historical documents written in Classical Chinese.

The group developed a pipeline that organizes page images, optionally enhances degraded pages, recognizes traditional Chinese text using PaddleOCR 3.7.0, reconstructs vertical reading order, restores punctuation using a Classical Chinese BERT model, and generates the required document-level outputs.

The final official corpus contains one `raw.txt` file and one `seg.tsv` file for each of the 33 document units covering 2,180 images. The workflow also includes resumption support, page-level traceability, output validation, and supplementary API-based comparison.

6.2 Remaining Limitations

The principal limitation is the absence of an expert-annotated ground-truth set. Internal confidence scores, structural checks, and API agreement cannot replace direct comparison with verified character transcriptions and sentence boundaries.

The OCR stage remains sensitive to degraded images, rare glyphs, mixed layouts, and possible Nôm characters. The coordinate-based reading-order rule may also require manual review for irregular pages.

The punctuation model may produce debatable boundaries and has limited context at chunk edges. The external APIs are not stable reference standards because their outputs can vary by prompt, model version, quota, and generative behavior.

Future work should therefore prioritize a manually verified evaluation subset. Such data would enable direct Character Error Rate and Boundary F1 evaluation against expert annotations. Further improvements may include OCR fine-tuning on historical materials, more advanced page-layout analysis, overlapping segmentation chunks, and controlled use of language-model APIs to flag uncertain passages for human review rather than automatically replacing the official text.